

HumanEval typicality metrics — qwen3.5-9b & gemma-4-31b-it (2026-06-16)

Settings: s2 = RankAlign; s4 = New+fsx+self-tc (FLORA-PMI); base = untrained qwen. Eval: basetyp = self-typ + base-typ; self = self-typ no-base; basetypneg = neg-typ + base-typ; neg = neg-typ no-base. Gen variant = tc (typicality-corrected). Metrics averaged over the 82 humaneval test tasks (gemma multi-s4-basetyp: 70, see /datastor1-full note).

Table 1: HumanEval typicality metrics (gen variant = tc), x100: gen ROC, val ROC, val acc, Pearson(gen,val).

Model	Data	Setting	Eval	gen ROC	val ROC	val acc	Pearson
gemma-4-31b-it (trained)	multi	s2	basetyp	90.68	89.95	76.46	64.86
gemma-4-31b-it (trained)	multi	s2	basetypneg	94.24	89.95	76.46	70.70
gemma-4-31b-it (trained)	multi	s2	neg	82.42	89.95	76.46	53.22
gemma-4-31b-it (trained)	multi	s2	self	85.63	89.95	76.46	58.34
gemma-4-31b-it (trained)	multi	s4	basetyp	90.13	94.37	82.64	69.02
gemma-4-31b-it (trained)	multi	s4	self	84.04	94.37	82.64	57.63
gemma-4-31b-it (trained)	upper	s2	basetyp	86.89	91.42	76.83	41.01
gemma-4-31b-it (trained)	upper	s2	basetypneg	88.22	91.42	76.83	55.54
gemma-4-31b-it (trained)	upper	s2	neg	86.26	91.42	76.83	59.93
gemma-4-31b-it (trained)	upper	s2	self	85.16	91.42	76.83	41.19
gemma-4-31b-it (trained)	upper	s4	basetyp	90.70	93.06	84.15	63.28
gemma-4-31b-it (trained)	upper	s4	self	84.33	93.06	84.15	52.54
qwen3.5-9b (trained)	multi	s2	basetyp	93.07	84.79	77.86	64.73
qwen3.5-9b (trained)	multi	s2	basetypneg	96.51	84.80	77.86	70.06
qwen3.5-9b (trained)	multi	s2	neg	80.32	84.79	77.86	75.23
qwen3.5-9b (trained)	multi	s2	self	90.21	84.76	77.86	61.32
qwen3.5-9b (trained)	multi	s4	basetyp	89.85	89.11	80.50	65.96
qwen3.5-9b (trained)	multi	s4	self	82.94	89.12	80.46	56.51
qwen3.5-9b (trained)	upper	s2	basetyp	87.25	83.28	74.31	55.32
qwen3.5-9b (trained)	upper	s2	basetypneg	90.22	83.23	74.31	69.48
qwen3.5-9b (trained)	upper	s2	neg	79.34	83.26	74.31	68.91
qwen3.5-9b (trained)	upper	s2	self	82.17	83.26	74.31	45.96
qwen3.5-9b (trained)	upper	s4	basetyp	83.23	86.33	76.45	58.86
qwen3.5-9b (trained)	upper	s4	self	78.58	86.30	76.53	51.49
qwen3.5-9b (BASE)	multi	base	neg	59.87	83.45	67.03	29.35
qwen3.5-9b (BASE)	multi	base	self	69.02	83.44	66.84	27.15
qwen3.5-9b (BASE)	upper	base	neg	65.73	81.70	65.28	30.42
qwen3.5-9b (BASE)	upper	base	self	69.07	81.72	65.28	26.48